

VSEPR-SIM Research Platform

§R Three Foundational Models

I. The 32-Bit Particle Identity Word

II. The Hourglass Convergence Model

III. The Lookglass Bidirectional Feedback Model

All three models operate on the Identity–State Decomposition framework introduced in §0.

Implementation files:

`coarse_grain/core/bit32_identity.hpp`
`coarse_grain/models/hourglass_model.hpp`
`coarse_grain/models/lookglass_model.hpp`

Status: Research-oriented active development

Contents

§1 The 32-Bit Particle Identity Word

1.1 Motivation

The §0 Identity–State Decomposition Framework defines a six-component particle identity vector

$$\mathbf{I}_i = \begin{bmatrix} Z_i \\ A_i \\ Q_i \\ \Sigma_i \\ \Lambda_i \\ \Theta_i \end{bmatrix} \quad (1)$$

where Z_i is atomic number, A_i is mass identity, Q_i is effective charge participation, Σ_i is structural role, Λ_i is stability class, and Θ_i is provenance memory.

In production simulations the identity vector is evaluated for every bead at every step. Storing it as six separate fields costs memory bandwidth and breaks cache locality. The 32-bit identity word packs all six fields into a single `uint32_t` that fits in a register, enabling branch-free scalar read-outs and SIMD-friendly bulk operations.

1.2 Bit-Field Layout

32-Bit Identity Word Layout					
31–24	23–16	15–8	7–5	4–3	2–0
Z_i [8]	A_i [8]	Q_i [8]	Σ_i [3]	Λ_i [2]	Θ_i [3]

Total width: $8 + 8 + 8 + 3 + 2 + 3 = \mathbf{32 \text{ bits exactly}}$.

1.3 Field Encodings

Field	Bits	Range	Encoding
Z_i	8	0–118	Atomic number; 0 = unassigned
A_i	8	0–255	Mass bucket; 0 = natural-abundance average
Q_i	8	$-32.0 \dots +31.75 e$	Signed s8.2 fixed-point, resolution 0.25 e
Σ_i	3	0–4	StructuralRole enum (5 values; 3 bits sufficient)
Λ_i	2	0–3	StabilityClass enum (4 values; 2 bits exact)
Θ_i	3	0–7	Provenance tag (8 provenance buckets)

1.3.1 Charge Encoding (Q_i)

Q_i uses signed 8-bit fixed-point with two fractional bits (s8.2), giving a resolution of 0.25 e and a range of $[-32.0, +31.75] e$. This is sufficient for all common ions (Th^{4+} , Fe^{3+} , organic anions) and avoids floating-point operations in the hot path.

The round-trip equation is:

$$q_{\text{fp}} = q_{\text{raw}} \times 0.25 e, \quad q_{\text{raw}} = \text{round}(q_{\text{fp}}/0.25 e) \quad (2)$$

1.3.2 Provenance Tag (Θ_i)

The full provenance hash lives in `Bead::parent_atom_indices`. The 3-bit tag is a coarse provenance bucket sufficient for routing decisions without dereferencing the parent record:

Tag	Meaning
0	Virgin / untracked
1	Mapped from atomistic parent
2	FIRE-relaxed minimum
3	CG-only (no atomistic parent)
4	Reaction product
5	Crystal seed
6	Foreign import
7	Hypothesis / mutated structure

1.4 Round-Trip Guarantee

Definition 1.1 (Round-Trip Consistency). A 32-bit identity word w is *round-trip consistent* if $\text{pack}(\text{unpack}(w)) = w$.

All valid inputs satisfy this property. The only invalid words are those with $\Sigma_i \in \{5, 6, 7\}$ (reserved), which are never produced by `pack_identity()`.

1.5 Implementation

The full implementation is in `coarse_grain/core/bit32_identity.hpp`. Key interface:

```

1 // Pack from individual fields
2 Identity32 id = pack_identity(Z, A, Q_fp, sigma, lambda, theta);
3
4 // Pack directly from a Bead
5 Identity32 id = pack_from_bead(bead, Z);
6
7 // Read individual fields (branch-free bit shifts)
8 uint8_t      z   = id.Z();
9 double       q   = id.Q();           // decoded to floating-point
10 StructuralRole sig = id.sigma();
11 StabilityClass lam = id.lambda();
12 ProvenanceTag th  = id.theta();
13
14 // Debug dump
15 std::cout << id.to_string() << '\n';
16 //      Identity32{ Z=6 A=0 Q=0      =Directional-Covalent      =Ambient-
17           Stable      =Mapped-Atomistic }
18
19 // Verify round-trip
20 assert(identity32_roundtrip_ok(id));

```

§2 The Hourglass Convergence Model

2.1 Concept

In standard molecular optimisation, a single trajectory descends a potential energy surface from an initial configuration to the nearest local minimum. This is efficient but blind: it finds *a* minimum, not necessarily the *best* minimum accessible from the current state.

The Hourglass Model addresses this by running a population of candidate configurations through a three-layer filter at each macro-step. The geometry of an hourglass is the natural metaphor:

Layer	Role
Wide mouth (H1)	Generate N_{cand} perturbed candidates
Narrow neck (H2)	Filter by stability, energy, and role weights
Wide base (H3)	Score and rank the survivors

2.2 Formal Specification

2.2.1 H1 — Mouth: Candidate Generation

Let $\mathbf{r}^{(0)}$ be the current bead configuration. Generate N_{cand} candidates:

$$\mathbf{r}^{(c)} = \mathbf{r}^{(0)} + \delta^{(c)}, \quad \delta_i^{(c)} \sim \mathcal{U}(-\delta_{\text{max}}, +\delta_{\text{max}})^3, \quad c = 1, \dots, N_{\text{cand}} \quad (3)$$

Uniform perturbations (not Gaussian) ensure the sampled volume in configuration space has a well-defined boundary at radius δ_{max} .

2.2.2 H2 — Neck: Three-Gate Filter

Each candidate passes three gates in sequence. Rejection at any gate eliminates the candidate immediately.

Gate 1 — Stability filter.

$$\bar{\Lambda}^{(c)} = \frac{1}{N} \sum_{i=1}^N \Lambda_i^{(c)} \geq \Lambda_{\text{min}} \quad (4)$$

Gate 2 — Energy filter.

$$\Delta E^{(c)} = E^{(c)} - E_{\text{basin}} \leq E_{\text{tol}}, \quad E_{\text{basin}} = \min_c E^{(c)} \quad (5)$$

Gate 3 — Role weight filter. Let $w_{\text{dom}}^{(i)}$ denote the dominant channel weight of bead i :

$$w_{\text{dom}}^{(i)} = \max(w_{\text{steric}}^{(i)}, w_{\text{elec}}^{(i)}, w_{\text{disp}}^{(i)}) \quad (6)$$

Then:

$$\frac{1}{N} \sum_{i=1}^N w_{\text{dom}}^{(i)} \geq w_{\text{min}} \quad (7)$$

2.2.3 H3 — Base: Composite Score

Surviving candidates are ranked by:

$$\mathcal{S}^{(c)} = w_E \cdot \frac{E_{\text{ref}} - E^{(c)}}{|E_{\text{ref}}| + \epsilon} + w_\Lambda \cdot \frac{\bar{\Lambda}^{(c)}}{3} + w_\Sigma \cdot \frac{1}{N} \sum_i w_{\text{dom}}^{(i)} \quad (8)$$

The top $k \leq N_{\text{out}}$ candidates are returned in descending score.

2.3 Parameters

Parameter	Symbol	Default	Meaning
N_{cand}	—	64	Candidate count
δ_{max}	—	0.5 Å	Perturbation radius
Λ_{min}	—	1 (Metastable)	Stability floor
E_{tol}	—	5.0 kcal/mol	Energy ceiling above basin
w_{min}	—	0.5	Role-weight floor
N_{out}	—	8	Maximum ranked survivors
w_E, w_Λ, w_Σ	—	1.0, 0.5, 0.3	Score weights

2.4 Implementation

The full implementation is in `coarse_grain/models/hourglass_model.hpp`.

```

1 HourglassParams params;
2 params.N_cand      = 128;
3 params.delta_max   = 0.3;    // tighter sampling
4 params.E_tol       = 3.0;    // stricter energy gate
5
6 HourglassResult result = run_hourglass(sys, params);
7
8 if (result.any_survived) {
9     const CandidateRecord* best = hourglass_best(result);
10    // Apply best configuration to system...
11 }
12
13 // Full audit: all candidates and gate verdicts
14 for (const auto& c : result.candidates) {
15     std::cout << "Candidate " << c.candidate_id
16               << "   E=" << c.energy
17               << "   survived=" << c.survived
18               << "   score=" << c.score << '\n';
19 }

```

2.4.1 Relationship to the Identity Vector

The Hourglass Model consumes Σ_i and Λ_i directly from each bead’s 32-bit identity word at Gate 1, Gate 3, and the H3 scoring function. It is therefore the first model in the platform that makes the Identity–State Decomposition *operationally load-bearing* rather than merely descriptive.

§3 The Lookglass Bidirectional Feedback Model

3.1 Concept

The 6+9 Seed-Bead Stepper is a *forward* pass: atomistic (seed) information propagates toward the coarse-grained (bead) layer. This is the natural direction of coarse-graining.

The Lookglass Model introduces the reverse: bead-layer observables are reflected back into the atomistic layer as soft corrections. The name is deliberately evocative — just as a lookglass (Lewis Carroll, *Through the Looking-Glass*) shows a mirror-reversed world, the Lookglass pass shows the atomistic layer a reversed image of its own coarse-grained representation.

Lookglass Architecture

Forward	[S1–S6] $\rightarrow$ [B1–B9] $\rightarrow$ SeedBeadStepRecord
Backward	[L1] pos correction $\leftarrow$ bead COM
	[L2] damping modulation $\leftarrow \bar{\eta}$
	[L3] force scale $\leftarrow$ role weights
	[L4] dt guard $\leftarrow \bar{\Lambda}$
	[L5] convergence mirror

3.2 Backward-Pass Unit Equations

3.2.1 L1 — Position Correction

The CG centroid for bead group i is $\mathbf{r}_i^{\text{bead}}$. The atomistic COM is $\mathbf{r}_i^{\text{seed}}$. The correction:

$$\Delta \mathbf{r}_i^{\text{seed}} = \alpha_{\text{pos}} (\mathbf{r}_i^{\text{bead}} - \mathbf{r}_i^{\text{seed}}) \quad (9)$$

$\alpha_{\text{pos}} \in [0, 1]$ is the position coupling constant. At $\alpha_{\text{pos}} = 0$ the backward pass is decoupled; at $\alpha_{\text{pos}} = 1$ atomistic positions are snapped to bead centroids.

3.2.2 L2 — Damping Modulation

The mean slow state $\bar{\eta}$ from the BEAD layer inflates the FIRE damping coefficient:

$$\gamma_{\text{eff}} = \gamma_0 \cdot (1 + \alpha_{\text{damp}} \cdot \bar{\eta}) \quad (10)$$

High $\bar{\eta}$ (ordered environment) $\rightarrow$ larger damping $\rightarrow$ slower but more stable relaxation.

3.2.3 L3 — Force Scale Correction

For each bead i with dominant role weight $w_{\text{dom}}^{(i)}$:

$$F_i^{\text{seed}} \leftarrow F_i^{\text{seed}} \cdot [1 + \alpha_{\text{force}} (w_{\text{dom}}^{(i)} - 1)] \quad (11)$$

Chemically specific beads ($w_{\text{dom}} > 1$) have their forces amplified; generic Mixed beads ($w_{\text{dom}} = 1$) are unchanged.

3.2.4 L4 — FIRE Step-Size Guard

Mean stability class $\bar{\Lambda} \in [0, 3]$ modulates the FIRE timestep ceiling:

$$\Delta t_{\text{max}} = \Delta t_0 \cdot \text{clamp} \left(1 - \alpha_{\Delta t} \frac{3 - \bar{\Lambda}}{3}, 0.1, 1.0 \right) \quad (12)$$

Unstable systems ($\bar{\Lambda} \approx 0$) reduce the timestep to at most 10% of Δt_0 ; bulk-lattice systems ($\bar{\Lambda} = 3$) run at the full timestep.

3.2.5 L5 — Convergence Mirror

The backward pass declares steady state when both:

$$\max_i |\Delta \mathbf{r}_i^{\text{feedback}}| < r_{\text{tol}} \quad \text{AND} \quad |\bar{\eta} - \bar{\eta}_{\text{prev}}| < \eta_{\text{tol}} \quad (13)$$

Once L5 is satisfied the backward pass is *frozen*: subsequent ticks skip the backward computation, recovering the pure 6+9 forward cost.

3.3 Lookglass Steady State

Definition 3.1 (Lookglass Steady State). The combined forward–backward system is in *Lookglass steady state* when the 6+9 Seed-Bead Stepper satisfies its own steady-state condition (§5.1) *and* the Lookglass backward pass satisfies equation (??).

The joint steady state is strictly stronger than either condition alone: it requires that the coarse-grained representation has converged *and* that its reflection no longer perturbs the atomistic layer.

3.4 Parameters

Parameter	Symbol	Default	Meaning
α_{pos}	—	0.05	Position coupling (L1)
α_{damp}	—	0.30	$\bar{\eta}$ damping inflation (L2)
γ_0	—	0.10	Base FIRE damping (L2)
α_{force}	—	0.20	Role-weight force scale (L3)
$\alpha_{\Delta t}$	—	0.50	Stability-class dt guard (L4)
Δt_0	—	5 fs	Base FIRE timestep (L4)
r_{tol}	—	0.001 Å	Feedback displacement tol (L5)
η_{tol}	—	10^{-4}	Slow-state convergence tol (L5)

3.5 Implementation

The full implementation is in `coarse_grain/models/lookglass_model.hpp`.

```

1 LookglassParams lg_params;
2 lg_params.alpha_pos = 0.05;
3 lg_params.alpha_force = 0.2;
4
5 LookglassRunRecord run_record;
6 run_record.record_history = true;
7
8 double eta_prev = 0.0;
9
10 for (uint64_t step = 0; step < max_steps; ++step) {
11     // Forward: 6+9 Seed-Bead step
12     SeedBeadStepRecord fwd = step_seed_bead(sys, sb_params);
13
14     // Backward: Lookglass reflection
15     LookglassStepRecord bwd =
16         step_lookglass(sys, fwd, lg_params, eta_prev);
17     run_record.ingest(bwd);
18
19     eta_prev = fwd.avg_eta;
20
21     if (fwd.steady_state && bwd.lookglass_steady) break;
22 }
23
24 std::cout << "Lookglass steady at step "
25             << run_record.steps_to_steady << '\n';
26 std::cout << "Peak feedback |dr| = "
27             << run_record.peak_pos_correction << " Å\n";

```


§4 Relationships Between the Three Models

The three models introduced in this document form a coherent stack:

Model	Consumes	Produces
32-Bit Word	$\mathbf{I}_i$ fields	Identity32 word
Hourglass	Bead population + Σ_i, Λ_i	Ranked candidate set
Lookglass	Forward step record + $\bar{\eta}, \bar{\Lambda}$	Corrected system state

4.1 Interaction Pattern

A production simulation loop using all three models proceeds as follows:

- 1. Pack identity words.** At initialisation (and after any structural event), pack each bead’s $\mathbf{I}_i$ into an **Identity32** word for fast lookup during force evaluation.
- 2. Forward pass.** Run the 6+9 Seed-Bead Stepper. Channel weights use Σ_i directly from the bead structs.
- 3. Backward pass.** Run `step_lookglass()` using the forward record. This corrects bead positions, adjusts FIRE damping and timestep, and checks L5.
- 4. Hourglass probe (periodic).** Every N_{hg} steps, run `run_hourglass()` to probe the local configuration space. If the best candidate scores substantially higher than the current state, accept the move.
- 5. Check joint steady state.** Halt when both the Seed-Bead Stepper and the Lookglass backward pass have converged.

4.2 Anti-Black-Box Guarantee

All three models honour the project-wide anti-black-box contract:

- `Identity32::to_string()` decodes every bit field by name.
- `CandidateRecord` exposes gate verdicts for every candidate.
- `LookglassStepRecord` exposes every backward-unit output.

Every decision made by these models can be reproduced, inspected, and reported without any hidden state.

Conclusion

This document introduced three foundational models that build directly on the §0 Identity–State Decomposition framework:

1. The **32-Bit Identity Word** compresses $\mathbf{I}_i$ into a single register-width integer, enabling branch-free field read-outs with no information loss.
2. The **Hourglass Model** operationalises Σ_i and Λ_i as load-bearing filter criteria, transforming the identity framework from a labelling system into a genuine constraint engine for structural convergence.
3. The **Lookglass Model** closes the loop between atomistic and coarse-grained layers, making the 6+9 Stepper bidirectional and enabling joint steady-state convergence.

Together they move VSEPR-SIM from a forward-only simulation pipeline toward a fully coupled, inspectable, and deterministic research platform.